SHANKAR M. DARUR

AI & Machine Learning Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI/ML graduate engineer with hands-on experience building Retrieval-Augmented Generation (RAG) systems, full-stack web platforms, and on-device/edge machine learning pipelines. Shipped a 20+ module production CMS as an industry intern, and independently designed a local-first security platform where all inference runs on-device. Founder of an early-stage café-booking startup with a live full-stack build.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, PHP, SQL, HTML, CSS

AI / ML: RAG, Hybrid Search (BM25 + Dense Vectors), Sentence-Transformers, NLP, Semantic Search

Frameworks: React, React Native (Expo), Next.js, FastAPI, Node.js, Express.js, Flask

Databases & Platforms: PostgreSQL, MySQL, MongoDB, ChromaDB, Redis, Prisma ORM, Docker, Git/GitHub, Render, Netlify, Vercel

PROFESSIONAL EXPERIENCE

Software Development Intern

Jan 2026 – Apr 2026

Agribid Private Limited — *Mumbai, India*

- Engineered a production CMS with 20+ modules for a live agritech platform; built a News Aggregation & RSS module auto-collecting agriculture news from multiple sources.
- Built responsive React frontends and Node.js/PHP backend modules integrated with MySQL via REST APIs; collaborated via Git on a live, in-production codebase.

Technologies Used: *Node.js, React, PHP, MySQL, REST API, Git*

Founder & Software Developer

Jan 2025 – Present

SeatSip — *Bangalore, India*

- Building a two-sided café-discovery and seat-booking platform: React Native (Expo) consumer app plus a Next.js admin/owner dashboard for managing seats, menus and bookings.
- Built the backend as a Node.js/Express/TypeScript REST API on Prisma ORM/PostgreSQL with Redis caching, JWT + Google OAuth, Razorpay payments, and a Socket.IO real-time delivery-tracking service.

Technologies Used: *Node.js, Express.js, TypeScript, Prisma ORM, PostgreSQL, Redis, React Native, Next.js, Socket.IO*

PROJECTS

SPECTER — AI Legal Assistant (RAG)

2025 – 2026

- Built a full-stack RAG prototype for Indian legal information: FastAPI backend, React/TypeScript frontend, ~59,600 indexed legal chunks in ChromaDB.
- Designed a hybrid retrieval pipeline (BM25 + dense-vector search, RRF fusion, cross-encoder reranker) — lifting Hit Rate by 37.5% and MRR by 21.2% over dense-only retrieval on a 50-question benchmark.
- Added a rule-based classifier for 10+ legal scenarios, OCR document intelligence, and a WhatsApp (Twilio) integration; containerised with Docker for Render/Netlify deployment.

Technologies Used: *React, FastAPI, ChromaDB, Sentence-Transformers, BM25, MongoDB, Tesseract OCR, Docker*

Specula — Local-First Application Security Platform

2026

- Unified SAST, DAST and network intrusion detection behind one API gateway and React dashboard — 7 Dockerised microservices, all ML inference on-device.
- Built a rule-based SAST engine (87% accuracy, 85% false-positive reduction) plus a fine-tuned CodeBERT classifier (99.6% validation accuracy); trained an XGBoost network-traffic classifier at 99.87% weighted F1 on NSL-KDD.

Technologies Used: *Python, FastAPI, Flask, React, Docker, CodeBERT, XGBoost, Isolation Forest*

EDUCATION

Bachelor of Technology (AI & Machine Learning) | 2022 – 2026

Basaveshwar Engineering College, Visvesvaraya Technological University, Karnataka, India

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification (Stanford Online) | DevOps, DataOps, MLOps (Duke University, Coursera)